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# Mesozoic palaeomagnetism in Vestfjella, Dronning Maud Land, East Antarctica

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**Summary.** Thermal and alternating field demagnetization of oriented samples of Mesozoic lava flows (200–230 Myr) and dykes (154–172 Myr) collected from seven nunataks within the Vestfjella region, Dronning Maud Land, revealed stable directions of magnetization of normal and reverse polarity. Directional distributions of both polarities define tight groups along the same palaeomagnetic axis. Depending on whether the regional westward dip of the lava flows originated prior to, or after the intrusive igneous phase, two or one, respectively, palaeomagnetic pole(s) can be estimated. Both poles, however, are in general accord with previous Mesozoic poles from East Antarctica.

## Introduction and geological summary

Prior to this study, stability-tested palaeomagnetic results from East Antarctica were limited to eight studies (McElhinny 1973), five of which represented investigations of Mesozoic rocks exposed along the Transantarctic Mountains. The present study adds further palaeomagnetic information to this period by describing results of palaeomagnetic investigations of Upper Triassic and Middle to Upper Jurassic igneous rocks from Vestfjella, Dronning Maud Land.

Vestfjella is situated on an ice-covered peninsula extending into the Weddell Sea and constitutes some of the northernmost exposures of the Transantarctic Mountains. It consists of a large number of nunataks, rising up to 400 m above the ice-cover, and extending in a NNE–SSW direction for some 130 km. The dominant rock type is subaerial basaltic lava flows, ranging in thickness from 1 to 25 m, dipping slightly to the WNW and building up a sequence of estimated stratigraphic thickness of around 2500 m (Furnes & Mitchell 1978). The lava flows are rather altered, reflected both by a greenish colour and the presence of an assemblage of secondary minerals such as quartz, calcite, prehnite and epidote. Field evidence suggests that this alteration, which apparently becomes more pronounced towards the base of the succession, is due to syn- and/or post-volcanic hydrothermal activity (Furnes & Mitchell 1978). The lava pile is cut by a large number of steeply dipping basaltic dykes 0.5–10 m in thickness. The strike is predominantly in a NW–SE direction, locally coinciding with the

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estimated fold axis of the lava flows. This suggests that the dyke intrusions may have caused the development of local low-amplitude folds (Hjelle & Winsnes 1972).

Apart from quartz and calcite fillings in the contact zones, the dykes are fresh-looking and 17 K-Ar determinations (Furnes & Mitchell 1978) give ages ranging between 156 and 172 Myr (Middle–Upper Jurassic). Four K-Ar ages of lava samples from the stratigraphically highest sections range between 200 and 231 Myr (Upper Triassic) of which the youngest ages are regarded as the most reliable ones (Furnes & Mitchell 1978).

## Field work

Samples for this study were collected by a field party visiting Vestfjella during the Norwegian Antarctic Expedition of 1976/77. A total of 200 oriented samples from 41 sites were collected from the named nunataks in Fig. 1. In addition to standard hand sampling, a portable drill was employed, by which more than half of the samples (126) were obtained. Orientation by both Sun and magnetic compass was accomplished on 154 samples, giving an arithmetic mean of the magnetic declination of 8.4° W ($\sigma = \pm 0.8°$). The small scatter of the mean suggests the absence of major local anomalies; hence the strike directions of samples oriented by magnetic compass only are corrected by this mean declination.

Attempts to trace magnetic polarity transitions within the stratigraphically lowermost section (Pagodromen) utilizing a portable flux-gate magnetometer, were not successful in that no sample gave any detectable deflection. The altered and badly cracked surface conditions put severe limitations on an effective sampling of the lava flows. Hence only 10 out of 42 sites represent lava flows.

## Magnetomineralogy

Properties of the magnetic minerals were investigated by ore microscopy and thermomagnetic analysis ($J_s - T$). The opaque mineralogy in one polished section from each site was studied

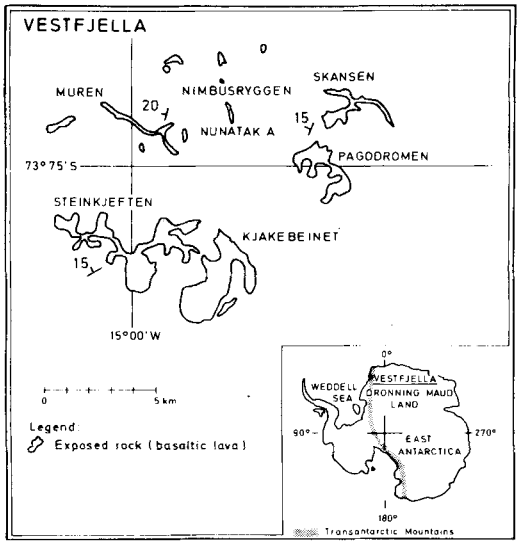



**Figure 1.** Maps showing the location of Vestfjella and positions of nunataks within the region of the present study. 'Nimbusryggen' and 'Nunatak A' are not official names.

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employing the magnetic colloid technique in order to detect and enhance the visibility of intragrain magnetic structures.

The lava flows from Pagodromen (Fig. 1) differ from all the other samples in the almost complete absence of visible opaque minerals even at the highest available magnification (× 2100). In reflected light the ground mass is brightly coloured in yellow, green, brown and red. Yellow-white structures are present which may represent relicts of titanomagnetite in oxidation classes II–VI. Low saturation magnetizations associated with predominantly hyperbolic thermomagnetic curves reflect major contributions from paramagnetic minerals although minor inflexion points around 560–580°C suggest the presence of small quantities of almost pure magnetite. Macrocrysts of prehnite coated with epidote reflect hydrothermal alterations at high pressures and temperatures (Ade-Hall, Palmer & Hubbard 1971), probably accounting for the dissolution of the deuteric crystallized magnetic minerals. The mineralogy in the four additional lava flows situated at higher stratigraphic levels (Kjakebeinet, Nunatak A and Muren) resemble the intrusive rocks with the exception that sulphide is absent.

The intrusive rocks generally have major mineralogical features in common. The opaque mineralogy is dominated by skeletal to euhedral titanomagnetite grains (5–100 $\mu$) in high temperature oxidation classes I–III (Ade-Hall *et al.* 1971). Low temperature alteration features such as maghemitization and granulation cover between 10–80 per cent of the grain surfaces. Non-magnetic cubic sulphides (< 10 $\mu$) in addition to slightly altered separate ilmenite rods amounts to less than 25 per cent of the opaque minerals.

Thermomagnetic analyses reveal a range of features reflecting different degrees of low-temperature alterations of the titanomagnetites. The $J_s - T$ curves are all irreversible with single initial Curie temperatures between 520 and 580°C, decreasing by 20–40°C after heating to 700°C, associated with either an increase or a decrease of the saturation magnetization. A feature resembling the 'kink' described by Ade-Hall *et al.* (1971), is present in about 10 per cent of the samples. Ade-Hall *et al.* believe this to be ascribed to a combination of high degree deuteric and hydrothermal alteration.

## Demagnetization experiments

One cylinder (19 × 19 mm) prepared from each sample was subjected to progressive alternating field ($\tilde{H}_{max}$ = 1000 Oe) and/or thermal demagnetization treatment. The direction and intensity of remanent magnetization was measured with a commercial slow-speed spinner magnetometer, and magnetic susceptibility was determined with a commercial low field inductance bridge. Results of the demagnetization treatment can be divided into three categories:

(1) Systematic directional trends terminating in stable end points defined as a cluster of at least three consecutive directions less than 5° of arc apart were found in 159 samples. Median destructive fields range from 150 to 400 Oe and are associated with a smooth decay of the remanent intensity. At the highest temperatures or peak alternating-fields the magnetization vector sometimes starts moving out from the stable end point groups in an erratic fashion. In order to retrieve palaeomagnetic information, great circles were fitted to the directional trends by a least-squares method (Halls 1976), and the intersection points of converging great circles satisfying adopted selection criteria (Halls 1976) were analysed. However, no unique intersection point deviating from the distribution of stable end points was found.

(2) Af-treatment proved unsuccessful in demagnetizing the remanence in samples from five dykes from Pagodromen, Steinkjeften and Kjakebeinet. The NRM directions remained stable even after treatment in fields of 1000 Oe, after which 45 per cent of the remanence

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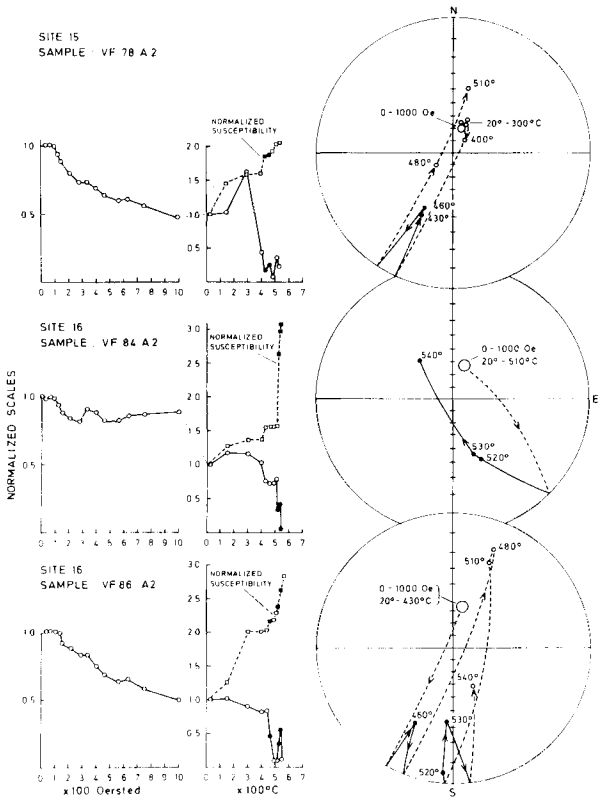



**Figure 2.** Composite demagnetization treatment (af and thermal) of three dyke specimens (Pagodromen) showing partial self-reversal (?) above 400° C. Stereographic plots of directions: open (filled) circles: up (down) pointing vectors. Normalized intensity curves: circles, normalized susceptibility curves: squares. Open (filled) symbols in accordance with corresponding directions.

intensity remained. Subsequent thermal demagnetization on top of the af treatment accomplished complete demagnetization of these specimens. On this procedure the remanent directions in samples from Kjakebeinet and Steinkjeften remained stable while the Pagodromen samples revealed a rather peculiar behaviour, demonstrated in Fig. 2. Above *c.* 400°C a rapid directional change takes place associated with a drop in the intensities and a pronounced increase in the susceptibilities measured after each heating step. The directional trends describe arcs of almost 180°. In some cases (*cf.* Fig. 2, VF78 and VF86), the directions, upon further heat treatment may turn back towards the initial directions. Although no stable antiparallel groups can be defined the directional behaviour, seen in conjunction with the distinct jump in the susceptibilities, tentatively suggest a case of partial self-reversal caused by magnetomineralogical changes (Creer 1971) induced during thermal treatment. Neither ore microscopy nor thermomagnetic studies could reveal significant differences between heated and unheated samples which contained titanomagnetite grains with more than 30 per cent granulation. Alternatively, these results could also reflect the presence of two opposed remanent directions, deviating significantly from being antiparallel. The pronounced increase in the susceptibility above 400°C could reflect the breakdown of mineral phases carrying the prevailing normal magnetization thus uncovering an overprint of reversed polarity.

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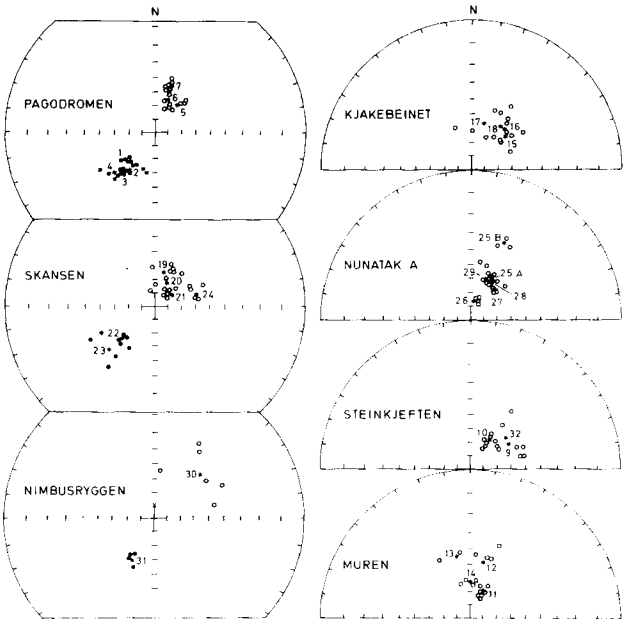



**Figure 3.** Stereographic plots of stable end points from each nunatak. Circles (squares): dykes (lava) specimens. Open (closed) symbols: up (down) pointing vectors. Site mean directions shown by star-symbol. Numbers refer to site numbers, *cf.* Table 1. Present dipole field direction indicated on lower left plot by 'X'.

(3) Systematic directional trends, encountered at the lower demagnetization levels in 20 samples, were succeeded by a scattered behaviour associated with a greater than 70 per cent reduction of the intensities. No stable end points were obtained and analyses of converging great circles fitted to the smoother fraction of the trends, was not successful in defining any unique intersection points.

## Palaeomagnetic directions

From demagnetization of 194 samples 174 gave stable end points. 32 of these directions (nine sites) were excluded from the final palaeomagnetic results adopting the following selection criteria:

(1) Each site should be represented by more than three stable end points and (2) site mean scatter ($\alpha_{95}$) should be less than 30°. Four sites were excluded on the grounds given in (1) and five due to (2).

Fig. 3 shows the remaining 142 stable directions plotted separately for each nunatak. Site mean directions and relevant statistical parameters are presented in Table 1. The directions, deviating significantly from the present geomagnetic field axis, are clustered in two almost antiparallel distributions. The normal polarity group has the highest concentration of results and appears to have a slightly steeper inclination than the reversed one.

Apart from large differences in the within site scatter, attributed to a combination of orientation effects and unsuccessful removal of deviating secondary components, only site 25 (Nunatak A) exhibits a somewhat anomalous distribution of stable directions. Collected across a 3-m thick dyke, the seven samples define two significantly different groups, 25A

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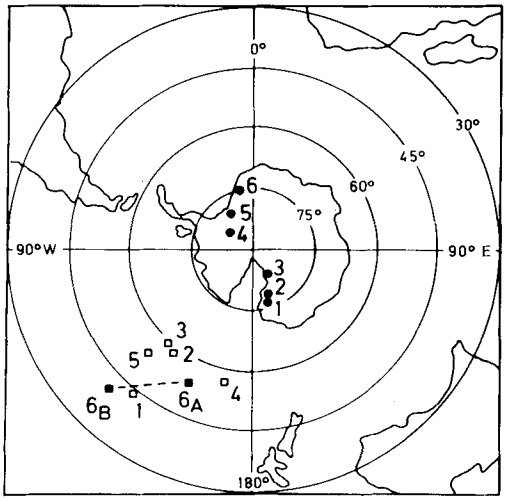



**Figure 4.** Mesozoic palaeomagnetic pole positions for East Antarctica (squares) and the corresponding sampling localities along the Transantarctic Mountains (circles). 1: Ferrar dolerite from Wright and Victoria Dry Valley (Bull & Irving 1960; Bull *et al.* 1962); 2: Ferrar dolerite from Ferrar Glacier (Turnbull 1959); 3: Ferrar dolerite from Beardmore Glacier (Briden & Oliver 1963); 4: Dufek intrusion (Beck 1972); 5: Ferrar dolerite from the Theron Mountains and Whichaway Nunataks (Blundell & Stephenson 1959; Blundell 1966); 6: Vestfjella, this study, $6_A$: no tectonic correction, $6_B$: corrected for 15° dip WNW.

and 25B (cf. Fig. 3). The former (four samples) coincides with the general normal distributions while the latter, a tightly clustered group of three directions, has a dip some 20° shallower than the former. The deviating distribution consists of samples collected from within 40 cm of the western contact, as opposed to the more central positions of site 25A samples, suggesting that the more rapid cooling rate of the contact zone may have provided the necessary conditions for inducing a spot reading of the geomagnetic field during the initial cooling of the igneous body.

The oldest lavas, sampled from the base of Pagodromen (sites 1–4), all have reversed polarity as opposed to the stratigraphically younger lava flows (sites 13, 14 and 26). The absence of both sedimentary and erosional surfaces within the lava pile (Furnes & Mitchell 1978) reflect a rather continuous emplacement of the lava flows during which at least one reversal of the geomagnetic field took place. The palaeomagnetic polarity time scale for the time interval in question (200–300 Myr) includes some 17 episodes of reversed polarity (McElhinny & Burek 1971), allowing for a rather rapid accumulation of the lava succession.

Within the time range for the intrusive igneous activity (156–172 Myr), only one reversed epoch (Mateke zone, around 170 Myr) has been recognized (McElhinny & Burek 1971), implying that the reversely polarized dykes were intruded during the early stage of the intrusive activity.

## Pole positions and discussion

Palaeomagnetic site mean directions presented in Table 1 and shown in the plots of Fig. 3 have not been subjected to any tectonic correction. Separate mean directions for lavas and dykes, irrespective of polarity, are presented in Table 2. The almost coinciding directions

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**Table 1.** Palaeomagnetic results.

| NUNATAK | SITE | LAVA/DYKE | N | $D_m$ | $I_m$ | $\kappa$ | $\alpha_{95}$ |
|---|---|---|---|---|---|---|---|
| PAGADROMEN | 1 | L | 4 | 224.2 | 60.3 | 592.1 | 3.8 |
| | 2 | L | 5 | 213.6 | 56.7 | 86.6 | 8.3 |
| | 3 | L | 5 | 218.8 | 52.4 | 53.6 | 10.6 |
| | 4 | L | 4 | 224.0 | 48.3 | 308.7 | 5.2 |
| | 5 | D | 5 | 39.9 | - 63.9 | 179.6 | 5.7 |
| | 6 | D | 5 | 20.8 | - 64.7 | 155.8 | 6.2 |
| | 7 | D | 5 | 16.5 | - 54.6 | 462.9 | 3.6 |
| SKANSEN | 19 | D | 5 | 15.2 | - 63.8 | 50.5 | 11.4 |
| | 20 | D | 5 | 27.0 | - 69.8 | 102.9 | 7.6 |
| | 21 | D | 5 | 44.6 | - 73.3 | 80.9 | 8.6 |
| | 22 | D | 4 | 224.4 | 56.3 | 585.8 | 3.8 |
| | 23 | D | 5 | 226.7 | 44.8 | 41.1 | 12.1 |
| | 24 | D | 5 | 74.0 | - 58.5 | 193.5 | 5.5 |
| KJAKEBEINET | 15 | D | 3 | 46.5 | - 55.6 | 61.6 | 15.8 |
| | 16 | D | 4 | 41.3 | - 53.8 | 37.7 | 15.2 |
| | 17 | D | 4 | 17.3 | - 55.1 | 16.9 | 23.1 |
| | 18 | L | 4 | 35.4 | - 52.2 | 86.2 | 9.9 |
| NUNATAK A | 25A | D | 4 | 24.1 | - 52.0 | 66.8 | 11.3 |
| | 26 | L | 4 | 21.0 | - 74.3 | 1835.7 | 2.2 |
| | 27 | D | 5 | 38.3 | - 62.1 | 842.3 | 2.6 |
| | 28 | D | 5 | 25.6 | - 57.8 | 1424.2 | 2.0 |
| | 29 | D | 5 | 25.1 | - 57.7 | 398.2 | 3.8 |
| STEINKJEFTEN | 8 | D | 3 | 38.5 | - 68.3 | 194.0 | 8.9 |
| | 9 | D | 4 | 57.8 | - 55.9 | 47.2 | 13.5 |
| | 31 | D | 4 | 11.0 | - 47.4 | 13.9 | 25.6 |
| | 10 | D | 3 | 32.2 | - 63.3 | 369.6 | 6.4 |
| NIMBUSRYGGEN | 30 | D | 6 | 47.4 | - 44.2 | 15.4 | 17.6 |
| | 31 | D | 4 | 208.5 | 56.3 | 330.7 | 5.1 |
| MUREN | 11 | D | 7 | 24.8 | - 69.2 | 367.1 | 3.2 |
| | 12 | L | 4 | 13.7 | - 48.2 | 27.0 | 17.7 |
| | 13 | D | 4 | 357.5 | - 39.0 | 17.4 | 22.7 |
| | 14 | L | 3 | 2.6 | - 63.2 | 360.3 | 6.5 |

N: number of samples within each site.
$D_m$: mean value of declination.
$I_m$: mean value of inclination.
$\kappa$: precision parameter.
$\alpha_{95}$: radius of 95 per cent circle of confidence.

**Table 2.** Palaeomagnetic mean directions and pole positions.

| | N | $\kappa$ | $\alpha_{95}$ | $D_m$ | $I_m$ | $\varphi$ | $\lambda$ | m | p |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Without tectonic correction | | | | | |
| Lavas | 7 | 41.55 | $9.47^\circ$ | $29.6^\circ$ | $-58.5^\circ$ | | | | |
| | | | | | | $54.4^\circ$S | $152.4^\circ$W | | |
| Dykes | 25 | 46.12 | $4.31^\circ$ | $31.6^\circ$ | $-58.9^\circ$ | | | | |
| | | | | Corrected | | | | | |
| Combined | 32 | 46.55 | $3.76^\circ$ | $50.9^\circ$ | $-52.0^\circ$ | $41.8^\circ$S | $133.5^\circ$W | $4.2^\circ$ | $5.6^\circ$ |

N: number of sites.
$\kappa$, $\alpha_{95}$, $D_m$, $I_m$: as described in Table 1.
$\varphi$, $\lambda$: latitude, longitude of palaeomagnetic pole.
m, p: semi-axes of ellipse of 95 per cent confidence of the pole.

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suggests an origin of the regional dip that post-dates the intrusive activity. The onset of sea floor spreading off Dronning Maud Land around 150 Myr ago (Bergh 1977) post-dates the intrusive episode in Vestfjella, and may thus represent the tectonic process resulting in the regional inclination of the lava flows. The fairly low precision (high $\alpha_{95}$) of the palaeomagnetic results from the lava sites, compared to that of the dykes, can probably be accounted for by the influence of local folds within the lavas, which originated during the intrusive episode (Hjelle & Winsnes 1972), and by the unsuccessful removal of magnetic components of opposite polarity.

Although the tectonic relationship between the nunataks are rather uncertain and the slight dip of the lava flows varies within each region, an overall bedding correction for a dip of 15° WNW is considered to be representative. The combined mean direction of all sites, corrected for tectonism, and the corresponding pole positions are presented in Table 2.

Previous Mesozoic palaeomagnetic poles from East Antarctica are plotted in Fig. 4 in addition to the pole positions from Vestfjella; $6_A$ and $6_B$; prior to and after tectonic correction respectively. Both poles are seen to fall within the general group of the previous poles. However, the uncorrected pole $6_A$, (*cf.* Table 2) is in perfect agreement with the suggested best estimate of the mean Mesozoic pole for East Antarctica (54° S, 154° W; Beck 1972). This could either be coincidental, or imply that the tilting of the lava pile pre-dates the intrusive phase. In this latter situation tilt correction should be applied to the lava rocks only. Considering the almost coinciding mean directions of the lavas and dykes, the corrected and uncorrected pole positions can be regarded as representative of the lavas and the dykes respectively.

Although the scatter between the pole positions in Fig. 4 can be accounted for by the expected behaviour of poles within the quasistatic interval for Gondwanaland (Briden 1967) covering the time span in question, the lower latitude of the tilt corrected, oldest pole, $6_B$, is tentatively suggested to reflect the trend of a general decrease in latitude of palaeomagnetic poles with increasing age. Thus, the two distinctly different pole positions $6_A$ (dykes) and $6_B$ (lavas), would represent Middle/Upper Jurassic and Upper Triassic poles respectively. In conclusion it should be stressed that the validity of the uncorrected pole, $6_A$, depend on an origin of the regional dip pre-dating the intrusive phase.

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